

Master of Computer Application (MCA) (Two Yrs Programme) - 3rd Semester (Batch 2024-26) (CBEGS)

(2225)

Paper: CSL-5038 : Advanced Software Engineering

Time allowed: 3 hrs.

Max. Marks: 70

Note: Attempt *five* Questions in all by selecting one question from each section. Fifth question may be attempted from any section.

Section A

1. a) Define the role of Quality Plan and Project Monitoring Plan in Software Project Planning. (7)
- b) What are the major problems faced by conventional software developers during software development? Discuss few Metrics used to improve conventional software development process. (7)
2. a) Explain major principals of Modern Software Management. (7)
- b) Discuss various techniques used to reduce size and improve process in order to improve software economics. (7)

Section B

3. a) What are the benefits of Software Reengineering? Define Horseshoe model for software reengineering. (7)
- b) Explain the concept of Program Structure Improvement and Program Modularization in re-engineering the existing system? (7)
4. a) What do you understand by Data Migration? Explain different activities involved in Data Migration Process. (7)
- b) Explain the significance of Software Reuse in software development life cycle. (7)

Section C

5. a) Explain CK Metrics Suite to measure the object-oriented design. (7)
- b) Explain Method-based design metrics for measuring object-oriented design along with their threshold value. (7)
6. a) What are the additional concepts covered by Rumbaugh methodology as compare to Booch methodology for Object-oriented analysis and design. (7)
- b) Discuss Rational Unified Process for software development in detail. (7)

Section D

7. a) What do you understand by Microservices Architecture? Design microservice architecture for an e-commerce app. (7)
- b) Define the concept of DevOps. Why code management is essential in DevOps? (7)
8. Define RESTful services. How it can be deployed? Discuss security and privacy measures employed on RESTful services to handle sensitive data. (14)
